INDIA INNOVATES 2026

Team Name:
EARTH DOOM BOTS

Domain 1- Urban Solutions

Problem Statement- Hyper-Local AQI & Pollution Mitigation Dashboard

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PROBLEM STATEMENT

Hyper-Local AQI & Pollution Mitigation Dashboard: Build a ward-wise, real-time air quality intelligence system that goes beyond city averages. The platform must use machine learning to detect localized pollution sources (e.g., construction dust, biomass burning) and provide automated policy recommendations for administrators along with health advisories for citizens.

Solution :-

Autonomous mobile robot with modular filtration + ML-powered hyper-local intelligence dashboard

❖ **our project aim to solve**

- ❑ Mobile robot actively purifies air while collecting ward-wise AQI data
- ❑ ML models detect pollution sources (construction, biomass, traffic)
- ❑ Real-time dashboard with Composite Ward Score prioritization
- ❑ Automated policy recommendations + citizen health advisories

GAP :-

- **City Averages Hide Reality:** **No ward-wise AQI tracking**
- **Blind Pollution Sources:** **Can't detect localised causes like construction dust or biomass burning in real-time across 272 wards.**
- **No Automated Policy:** **Administrators rely on manual GRAP triggers instead of source-specific recommendations.**
- **Missing Citizen Health Alerts:** **Generic warnings fail to deliver ward-level advisories for vulnerable groups.**
- **No Robotics/Hardware Mitigation:** **Zero autonomous mobile systems exist to physically target and clean hyper-local pollution hotspots.**

SOLUTION

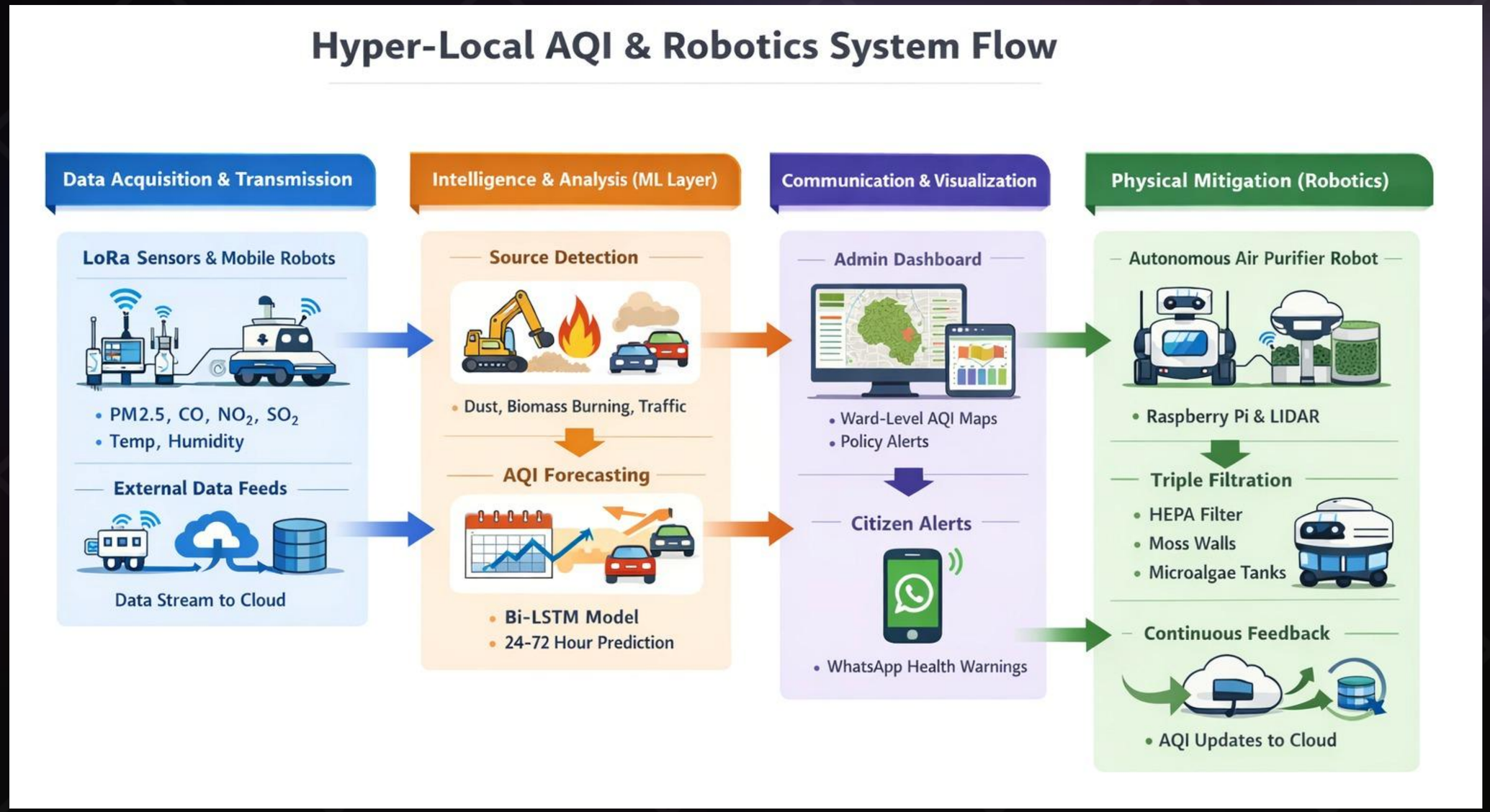
Core Approach:

- Robots auto-navigate to high-pollution wards using Raspberry Pi + LIDAR, deploy HEPA/algae filters (500 m^3/hr capacity), report back via IoT
- Dashboard shows ward-wise AQI maps; admins get simple alerts
- Citizens get ward-specific health alerts via WhatsApp
- **Predictive Analytics:** Bi-LSTM model forecasts 24-72hr ward AQI using weather + traffic data; Composite Score = (AQI × Pop Density × Health Vulnerability) prioritizes action.
- **Triple Filtration System:** HEPA (99% PM2.5 capture), moss walls (20% CO2 absorption), microalgae tanks (15% NOx removal)

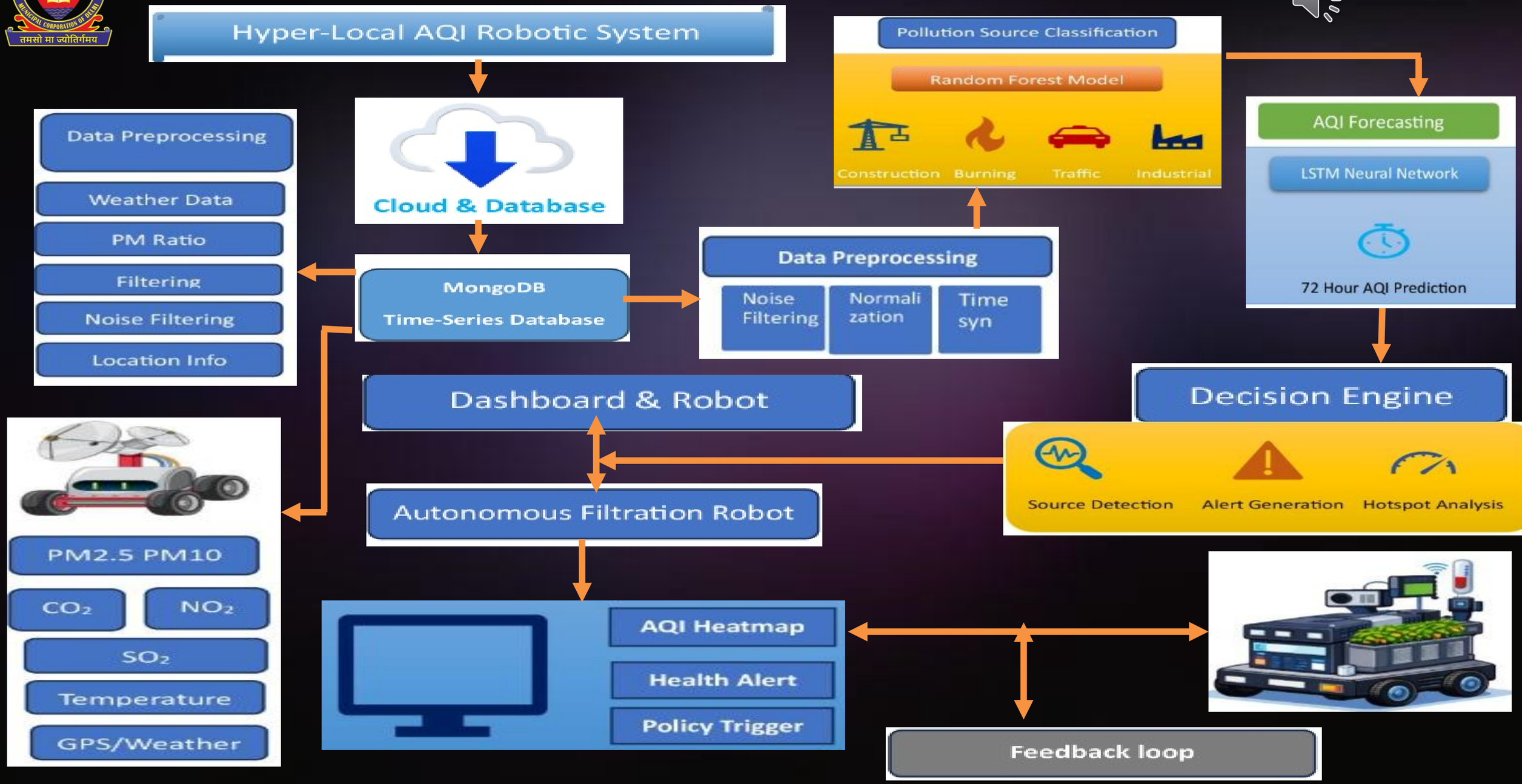
What Makes It Stand Out:

- First system combining monitoring + physical robot mitigation
- Targets exact pollution sources with autonomous deployment
- Uses existing DPCC APIs + MCD ward maps for immediate rollout
- Low-cost solar robots vs expensive static stations
- GRAP-compliant automated alerts for MCD/DPCC
- Pilot-ready for 5 worst wards (Narela, Mundka, Wazirpur)
- **Bottom Line:** Sensors → ML source ID → Robot cleaning → Actionable alerts

ARCHITECTURE



SYSTEM ARCHITECTURE AND WORKFLOW



TECHNOLOGY USED

Hardware & Software Stack

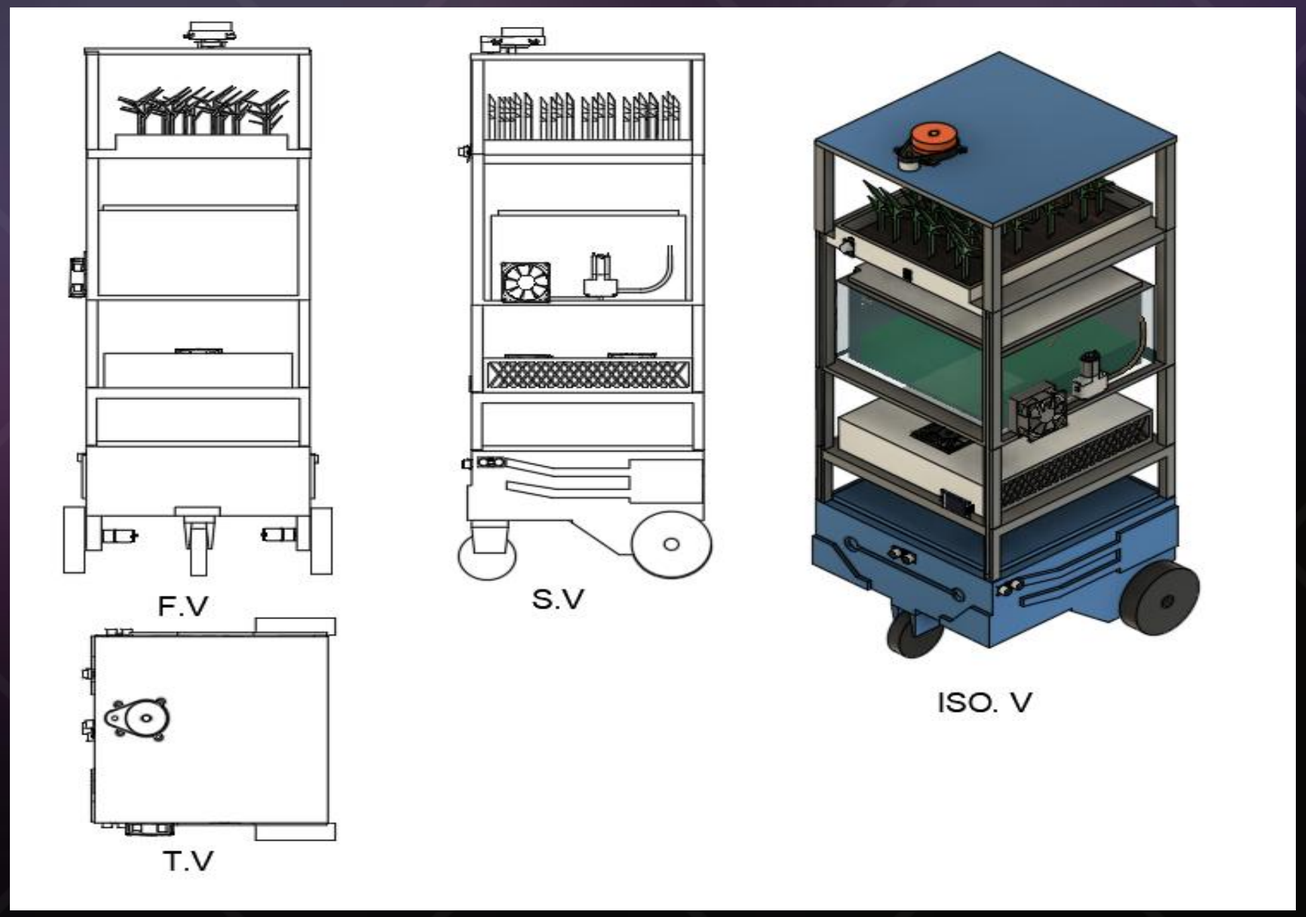
Hardware Components:

Raspberry Pi 4 (Robot brain: navigation, SLAM)
ESP32 (IoT gateway: sensor data collection)
Arduino Uno (Motor/fan/pump control)
MQ2/MQ135 Sensors (Gas detection)
DHT22 (Temp/Humidity)
LIDAR/L298N Motors (Robot mobility/filtration)

Software & Frameworks:

MQTT Protocol (Real-time data transmission)
TimescaleDB (Time-series ward data storage)
Python (ML pipeline: source detection, forecasting)
Node-RED (Dashboard flows)
WhatsApp Business API (Citizen alerts)
ROS (Robot Operating System for navigation)
Integrations: DPCC APIs, IMD weather feeds, MCD ward boundaries

MOBILE ROBOT CAD MODEL



FEATURE / USP

Feature	Our Solution	Existing Solution
Monitoring	Hyper-local, Ward-wise with mobile data collection	City wide averages only
Mitigation	Autonomous Robot Purification(500m3/hr)	Passive/static sensors, no cleaning
Intelligence	Ml source ID (24-72hr Forecasts)	Blind to localized cost
Delivery	WhatsApp Alert, Auto policy recs	Manual GRAP, Generic Warning
Cost	Low-cost solar, ROS/MQTT, Existing APIs	High-cost station

REFERENCES / LINKS

- https://cpcb.nic.in/-India's national pollution monitoring authority.
- https://dpcc.delhigovt.nic.in/-Delhi government agency for pollution regulation
- https://www.who.int/airpollution-Global health guidelines and research on air pollution.
- https://www.ros.org/-Open-source framework for robotics development.
- https://www.raspberrypi.org/documentation-Official documentation for Raspberry Pi hardware and projects.
- https://mausam.imd.gov.in/-India's official weather and climate data provider.
- https://mqtt.org/-Lightweight IoT messaging protocol.
- https://direct.mit.edu/neco/article-abstract/9/8/1735/6109/Long-Short-Term-Memory?redirectedFrom=fulltext-Foundational deep learning model for sequence prediction.
- https://link.springer.com/article/10.1023/A:1010933404324

THANK YOU